MODULE 4 ENDOCRINE ANATOMY

Endocrine Anatomy.

ENDOCRINE ANATOMY

Covers the endocrine glands, where each one sits in the body, and the gross anatomy of the pituitary, thyroid, parathyroid, adrenal, and pancreatic glands. Hormone action belongs to physiology and is not covered here.

BY THE END

1. Name the major endocrine glands and locate each one in the body.
2. Describe the gross anatomy of the pituitary gland and its connection to the hypothalamus.
3. Identify the parts of the thyroid, parathyroid, adrenal, and pancreatic glands.
4. Name organs that are not endocrine glands but still contain endocrine tissue.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Endocrine anatomy

drsrennie-stack.github.io/new-build-bio4-solano/endocrine-concept-videos.html

The Endocrine System, an Overview

The endocrine system is a scattered set of glands and tissues that signal the body by releasing hormones into the blood. Covers where those glands sit and how each one is built.

THE BASIC VOCABULARY

Structure	What it is
Endocrine system	The glands and tissues that release hormones into the blood to signal distant cells
Hormone	A chemical messenger released by an endocrine cell that travels in the blood to act on target cells
Endocrine gland	A ductless gland that secretes its product directly into the surrounding fluid and blood
Exocrine gland	By contrast, a gland that secretes through a duct onto a surface; not part of the endocrine system
Target cell	A cell that carries the receptor for a given hormone and therefore responds to it

The Major Endocrine Glands

The pure endocrine glands do nothing but make hormones. They are spread from the brain to the pelvis. Compare them by where each one sits.

WHERE EACH GLAND SITS

Gland	Location	Note
Hypothalamus	The floor and walls of the third ventricle, in the diencephalon of the brain	Links the nervous and endocrine systems and directs the pituitary

Gland	Location	Note
Pituitary gland	In the sella turcica of the sphenoid bone, hanging below the hypothalamus	The small gland the hypothalamus controls
Pineal gland	The posterior roof of the third ventricle, in the brain	A small, cone-shaped gland
Thyroid gland	The anterior neck, just below the larynx, wrapped around the trachea	The largest purely endocrine gland
Parathyroid glands	The posterior surface of the thyroid gland, usually four small glands	Tiny glands embedded behind the thyroid
Thymus	The mediastinum, behind the sternum	Large in childhood, shrinks after puberty
Adrenal glands	One capping the superior pole of each kidney	Also called the suprarenal glands
Pancreatic islets	Scattered through the pancreas, behind the stomach	The endocrine portion of the pancreas
Ovaries	The female pelvis, one on each side of the uterus	The female gonads
Testes	The scrotum	The male gonads

The Pituitary Gland

The pituitary gland, or hypophysis, is about the size of a pea. It sits in the sella turcica and hangs from the hypothalamus by a stalk. Its two lobes come from two different tissues.

THE PITUITARY AND ITS CONNECTIONS

Structure	What it is
Hypophysis	Another name for the pituitary gland
Sella turcica	The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary
Infundibulum	The slender stalk that connects the pituitary to the hypothalamus above it
Hypophyseal portal system	The set of blood vessels that carries hypothalamic signals directly to the anterior lobe

The two lobes

Compare the two lobes by the tissue they are made of and how each connects to the hypothalamus.

THE LOBES OF THE PITUITARY GLAND

Lobe	Other name	Tissue	Link to the hypothalamus
Anterior lobe	Adenohypophysis	Glandular epithelial tissue	The hypophyseal portal system, a set of blood vessels
Posterior lobe	Neurohypophysis	Nervous tissue, an extension of the hypothalamus itself	A tract of nerve fibers running down through the infundibulum

KEY POINT

The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels. The tissue tells you the link.

The Thyroid and Parathyroid Glands

The thyroid gland sits across the front of the trachea, and the four small parathyroid glands are embedded in its back surface.

The thyroid gland

THE THYROID, FROM THE GROSS GLAND TO THE FOLLICLE

Structure	What it is
Thyroid gland	The largest purely endocrine gland, two lateral lobes lying against the trachea below the larynx
Isthmus	The bridge of thyroid tissue, crossing the front of the trachea, that connects the two lobes
Thyroid follicles	The microscopic hollow spheres that make up the gland
Colloid	The protein-rich fluid stored in the center of each follicle
Follicular cells	The cells forming the wall of a follicle
Parafollicular cells	Also called C cells, scattered between the follicles

The parathyroid glands

THE PARATHYROID GLANDS AND THEIR CELLS

Structure	What it is
Parathyroid glands	Usually four small, round glands on the posterior surface of the thyroid lobes
Chief cells	The main hormone-producing cell of the parathyroid glands

The Adrenal Glands

An adrenal, or suprarenal, gland caps the top of each kidney. Each gland is really two glands in one: an outer cortex and an inner medulla, formed from different tissues.

THE TWO PARTS OF AN ADRENAL GLAND

Structure	What it is
Adrenal cortex	The outer region and the bulk of the gland, glandular tissue arranged in three layered zones
Adrenal medulla	The inner core, modified nervous tissue that belongs to the sympathetic nervous system

The cortex is built in three zones. Follow them in order, from the surface inward.

1. **Zona glomerulosa** the thin outer zone, just beneath the connective tissue capsule
2. **Zona fasciculata** the thick middle zone, the widest of the three, its cells in straight columns

3. **Zona reticularis** the inner zone, a branching network of cells against the medulla

The Pancreas and the Gonads

The pancreas and the gonads are mixed organs. Each does an endocrine job alongside another role, so only part of the organ is endocrine tissue.

THE MIXED ORGANS

Structure	What it is
Pancreas	A soft gland behind the stomach, both an exocrine and an endocrine organ
Pancreatic islets	Also called the islets of Langerhans, the small clusters of endocrine cells scattered through the pancreas
Acini	By contrast, the exocrine clusters of the pancreas that make digestive enzymes; not endocrine tissue
Ovaries	The paired female gonads in the pelvis, endocrine glands as well as reproductive organs
Testes	The paired male gonads in the scrotum, endocrine glands as well as reproductive organs

Organs with Endocrine Tissue

Beyond the pure endocrine glands, many organs whose main job is something else still carry pockets of endocrine cells. Compare them.

ORGANS THAT CONTAIN ENDOCRINE TISSUE

Organ	Endocrine tissue it contains
Hypothalamus	Neurosecretory cells; a neuroendocrine organ that controls the pituitary gland
Heart	Hormone-producing cells in the wall of the atria
Kidneys	Cells that release a hormone driving red blood cell production, plus an enzyme acting on blood pressure
Stomach and small intestine	Scattered endocrine cells in the lining of the digestive tract
Liver	Cells that contribute to a growth-related hormone and to blood pressure control
Skin	Cells that begin the production of vitamin D, which acts as a hormone
Adipose tissue	Fat cells that release a hormone signaling the body's energy stores
Placenta	A temporary organ of pregnancy that releases hormones supporting the pregnancy

WHERE A GLAND SITS SHAPES ITS DISEASE

The **pituitary gland** is boxed inside the bony **sella turcica**, so a pituitary tumor has nowhere to expand but upward, where it presses on the optic chiasm. A loss of peripheral vision can be the first sign that a gland the size of a pea has enlarged.

The **thyroid gland** lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny **parathyroid glands** on its back surface must be carefully spared.

Each **adrenal gland** rests on the superior pole of a kidney, which is why imaging ordered for the kidney also captures the adrenal, and why an adrenal mass is sometimes found by accident.